

AIoT UPPER-ARM WEARABLE

Nova-Band

An AIoT-driven upper-arm wearable for real-time physiological monitoring and running performance

TEAM NOVA-BAND

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PPG + IMU

EDGE FFT · ESP32

EARLY WARNING

IN COLLABORATION WITH



UPPER-ARM FIT · PPG + IMU · AIoT

What we will cover

Five chapters, from the running-injury problem to evidence and market.

**01**

The running problem

Sedentary lifestyles, overtraining and unreliable wrist data.

**02**

The Nova-Band system

Upper-arm fit, PPG + IMU, edge FFT and an AI early warning.

**03**

Product & hardware

Module anatomy, 3D digital twin and specifications.

**04**

Software & AI

The runner app, Readiness Score and the community dashboard.

**05**

Evidence & market

MAPE and SUS results, market size and projection.

Background

Physical inactivity is a public-health problem — and running is where many people start.



37.4%

of Indonesians aged ≥ 10 are not active enough

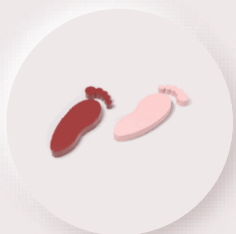
Below the recommended level of physical activity. SKI 2023



3 Million+

deaths a year linked to physical inactivity

One of the leading global risk factors for premature death. WHO 2020



1–42%

MAPE of consumer wearables in step counting

Accuracy varies widely between devices. Al-Marroof et al. 2024



15–44%

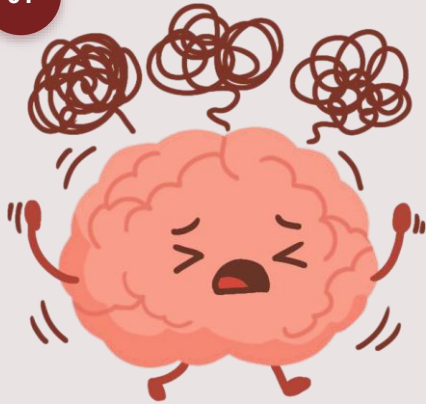
MAPE in energy / calorie estimation

Even wider error on calories burned. Al-Marroof et al. 2024

Meet the **problem**

Four ways today's running culture — and wrist wearables — fail runners.

01



Worried-well anxiety

Excessive anxiety about digital metrics on the watch screen.

02



Physical harm

Runners ignore pain and extreme fatigue to hit on-screen targets.

03



Overtraining risk

Pushing beyond physical limits, driven by FoMO and lifestyle pressure.

04

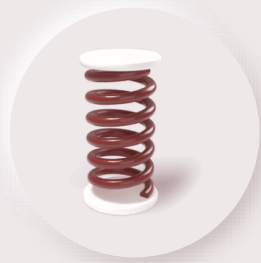


Data inaccuracy

Wrist sensors lose accuracy exactly when running gets intense.

Trends in runner injuries

The injury pattern has shifted toward heavier soft-tissue and tendon damage.



Running is booming — so is overtraining

Among urban runners, lifestyle pressure, commodification and Fear of Missing Out push people past their physical limits. Injuries used to be dominated by kneecap pain; they now shift to heavier soft tissue and tendons.

Two main predictors excess mileage · previous injury history

1

Achilles tendinopathy

Tendon overload behind the ankle

2

Calf strain

Gastrocnemius injury under repeated push-off

3

Hamstring & quadriceps strain

Thigh muscles overstretched without adequate warm-up

4

Iliotibial Band Syndrome (ITBS)

Outer-knee pain linked to high training load

5

Knee meniscus tear

Cartilage damage from accumulated impact

Research questions

Two problems formulated, two objectives set.

PROBLEM FORMULATION

1

How do motion artifacts from arm swing affect the accuracy of heart-rate and motion data in wrist-worn wearables used by runners?



2

How can Nova-Band be designed and tested to extract an accurate Readiness Score that helps prevent soft-tissue injury and cardiovascular collapse?



RESEARCH OBJECTIVES

1

Design Nova-Band as an upper-arm wearable that mechanically stabilises the sensors to minimise motion artifacts and produce accurate PPG and IMU data.



2

Evaluate its effectiveness through MAPE accuracy testing and SUS usability testing in extracting a reliable Readiness Score.



State of the **art**

What the literature shows — and the gap Nova-Band closes.

Siradj
2016

Wearables help runners track physical activity practically and independently for self-motivation.



Gap Consumer accuracy is low: MAPE 1–42% for steps, 15–44% for calories.

Al-Maroofof et al.
2024

Smartwatches enable continuous tracking of physiological metrics and self-monitoring interventions.



Gap Reliability and validity vary across devices, especially at high intensity.

Moghaddam et al.
2025

Placement studies show wrist-worn devices are practical for everyday heart-rate monitoring.



Gap Wrist PPG is highly vulnerable to motion artifacts during intense running.

McGrath & Doyle et al.
2023

IMU sensors with machine learning estimate ground reaction force and running kinematics precisely.



Gap Largely lab-grade — not yet integrated for real-time daily use by runners.



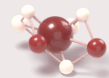
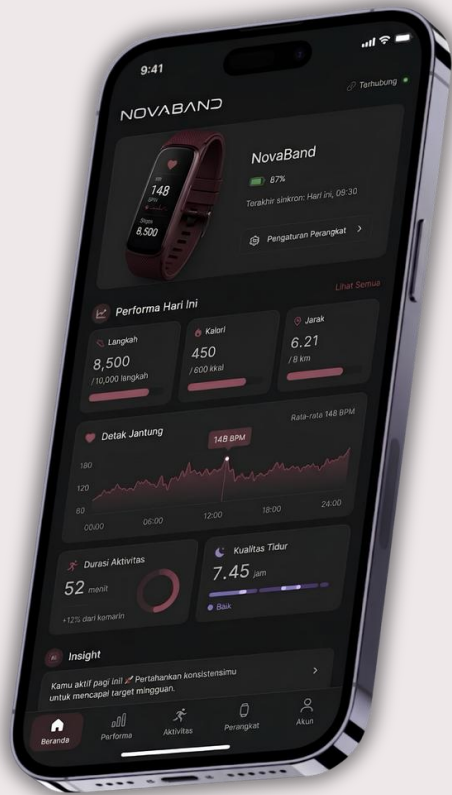
Nova-Band's answer ✓ Upper-arm placement reduces motion artifacts

✓ PPG, IMU and ESP32 integrated: cardio + biomechanics

✓ IoT dashboard alerts organisers early to prevent overtraining

The solution

Four layers of protection in one band.



AI & IoT Synergy

Sensor readings from the band (IoT) are processed continuously by AI into physiological insight.



Shared Safety Net

A dual-layer system: alerts reach the runner and the supervisor at the same time to prevent collapse.



Biological Early Warning

Proactive, instant alerts the moment a runner starts pushing past the body's safe limits.



Soft Network Load

IMU kinematic tracking of mechanical stress on muscles and tendons — against Achilles tendinopathy & ITBS.

Our novelty

Three things a wrist wearable does not do together.

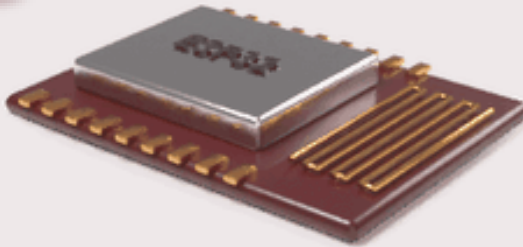
1



Upper-arm fit

Worn away from the arm's largest swing, so the optics stay pressed to the skin and motion artifacts drop — a mechanical fix first.

2



Edge FFT on the ESP32

A Fast Fourier Transform on the device separates heart-rate frequencies from cadence noise before any data leaves the band.

3



Shared safety net

Data syncs to a community dashboard, so an event supervisor sees an alert at the same moment the runner does.

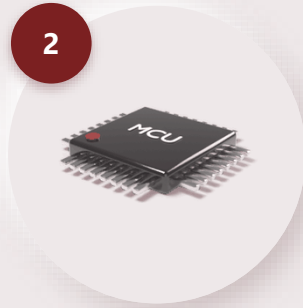
How Nova-Band **protects** runners

From a beam of light on the skin to an alert on the supervisor's screen.



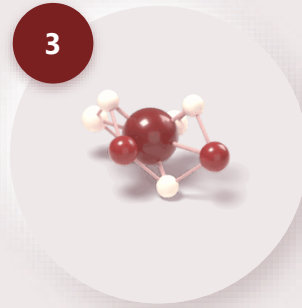
Acquire

PPG and IMU sampled together on the upper arm.



Filter

FFT on the ESP32 strips cadence noise from the signal.



Analyse

AI turns clean signals into a Readiness Score.



Notify

Bluetooth streams metrics to the runner's app.



Protect

The IoT dashboard watches the whole group.



An alert reaches two people at once

When a runner crosses a safe limit, the warning goes to the runner and to the event supervisor simultaneously — the Shared Safety Net.

Hardware operation (PPG)

Photoplethysmography: every heartbeat changes how much light the skin gives back.



LED Sensor

Green LEDs shine into the skin of the upper arm.



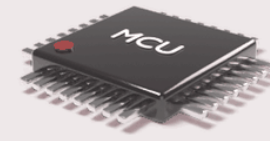
Blood Absorbs Light

Blood volume changes with each beat, absorbing light.



Photodetector

Captures the light reflected back from the skin.



Microcontroller

The ESP32 filters the signal with an on-device FFT.



Heart Beats

Becomes real-time BPM and feeds the Readiness Score.

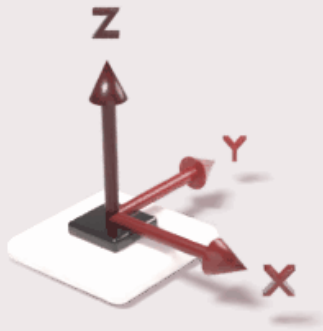
What the band measures

An LED shines into the skin; with every heartbeat the blood volume below changes, and so does the light that returns to the photodetector. The ESP32 filters that signal and turns it into heart rate.

2.5–3.5% MAPE

Hardware operation (IMU)

Three motion sensors that watch for emergencies.



Accelerometer

FALLS & COLLAPSE

Detects drastic changes in motion speed to identify sudden falls or physical collapse.



Gyroscope

POSTURE & BALANCE

Monitors body posture and tilt to detect loss of balance, dizziness or unsteady movement.



Magnetometer

ORIENTATION

Determines precise body orientation to assist spatial tracking in emergency situations.

Meet our **product**

An upper-arm band that keeps its sensors still while you run.



Upper-arm fit

Sits away from the arm's largest swing — the optics stay in contact at running pace.



PPG + IMU in one module

Heart rate and body movement captured together, filtered on the ESP32.



Built to be worn

Precision 3D-printed, sweat-resistant casing on a porous modular elastic strap.



NOVA-BAND · PRODUCT

Product anatomy

Five layers inside the module, from the optics to the skin.



Blender model · internal layout illustrative

1

Optical window

Two green LEDs, an IR diode and a photodetector

2

Casing

Precision 3D-printed polymer, sweat-resistant

3

Li-Po battery

High-density cell for long endurance runs

4

Mini PCB

ESP32, IMU and step-down regulator

5

Sensor pod

Skin side — keeps the optics in contact

3D digital twin

A Blender model rebuilt from the product images — and reused everywhere.



PRODUCT IMAGE



BLENDER 3D

One model, every medium

- ✓ Interactive 3D viewer on the website
- ✓ 46-asset icon & render library
- ✓ Animated loops in this deck
- ✓ 1920 × 1080 showreel video

Modelled from the product images: curved pill module, framed optical window, woven strap with stitched keeper.

Upper-arm fit

Why the band lives on the upper arm, and how it is used.



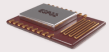
Why the upper arm?

The wrist swings the most while running, and that swing bleeds into the optical signal. The upper arm moves less, so the sensor stays in contact.

- 1 Strap the band on the upper arm, module facing out.
- 2 Pair it with the Nova-Band app over Bluetooth.
- 3 Run — heart rate, cadence and load stream in real time.
- 4 Alerts reach you and your supervisor if a limit is crossed.

Specifications

What is inside the prototype.



MICROCONTROLLER

ESP32 on a custom mini PCB



MOTION SENSOR

IMU — accelerometer, gyroscope, magnetometer



CASING

Precision 3D-printed polymer



POWER

Li-Po + step-down regulator



PHYSIOLOGICAL SENSOR

Optical PPG



PROCESSING

On-device FFT, C++ firmware



STRAP

Porous modular elastic



CONNECTIVITY

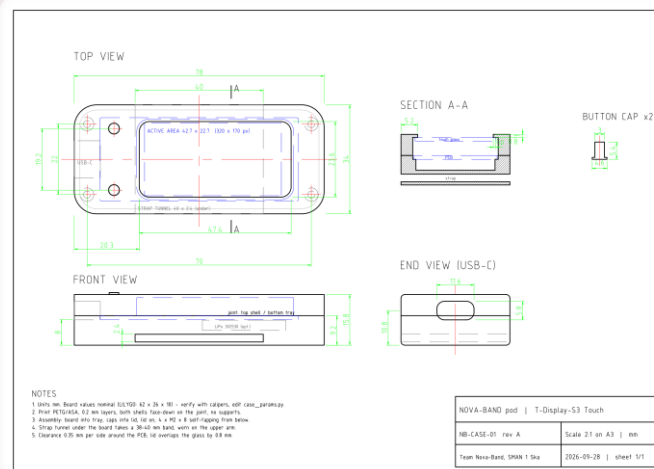
Bluetooth → app → IoT dashboard

Working prototype

A LILYGO T-Display-S3 Touch in a 3D-printed pod — sketched in LibreCAD, solid-modelled in FreeCAD.



PRINTED POD · PETG



LibreCAD sketch · NB-CASE-01

Top, front, end and section views, A3 at 2:1. Board values are nominal — measured before printing.

DXF · STEP · STL

MCU

ESP32-S3R8 · 240 MHz dual core

MEMORY

16 MB flash · 8 MB PSRAM

DISPLAY

1.9" IPS 320×170 · touch

CASE

78 × 34 × 15.8 mm · 2 parts + caps

STRAP

38–40 mm band, under-tunnel

LINK

Bluetooth LE · USB-C

On-device interface

Firmware for the band's own screen — Blender assets, depth of field, 60 fps on both CPU cores.



Blender splash

A rack-focus intro rendered in Blender, played from flash.



Depth of field

Gym bokeh far, glass cards mid, blurred specks near — three parallax layers.



60 frames per second

Each core draws half the frame; DMA sends it to the ST7789.



Touch & buttons

Swipe with spring physics; long-press to start or pause a run.

Connected to the **app**

Bluetooth LE or USB, one JSON protocol — and the web app installs like a native app (PWA).



NovaBand · pairing page

⇄ BLUETOOTH LE

Web Bluetooth · Android, Windows, macOS

⇄ USB-C

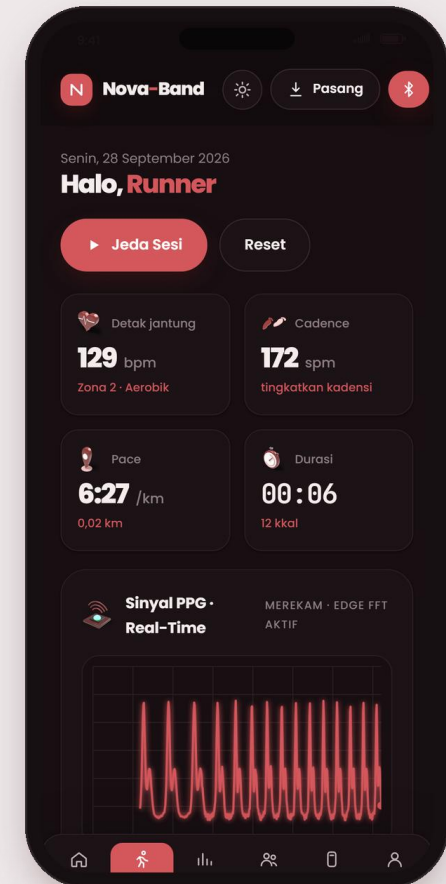
Web Serial · desktop Chrome / Edge

Telemetry 1 Hz · PPG wave 50 Hz

Standard Heart Rate service 0x180D

App → band: run, alerts, coach messages

The QR code on the band opens **novaband-id.web.app** — installable, works offline.



In the gym



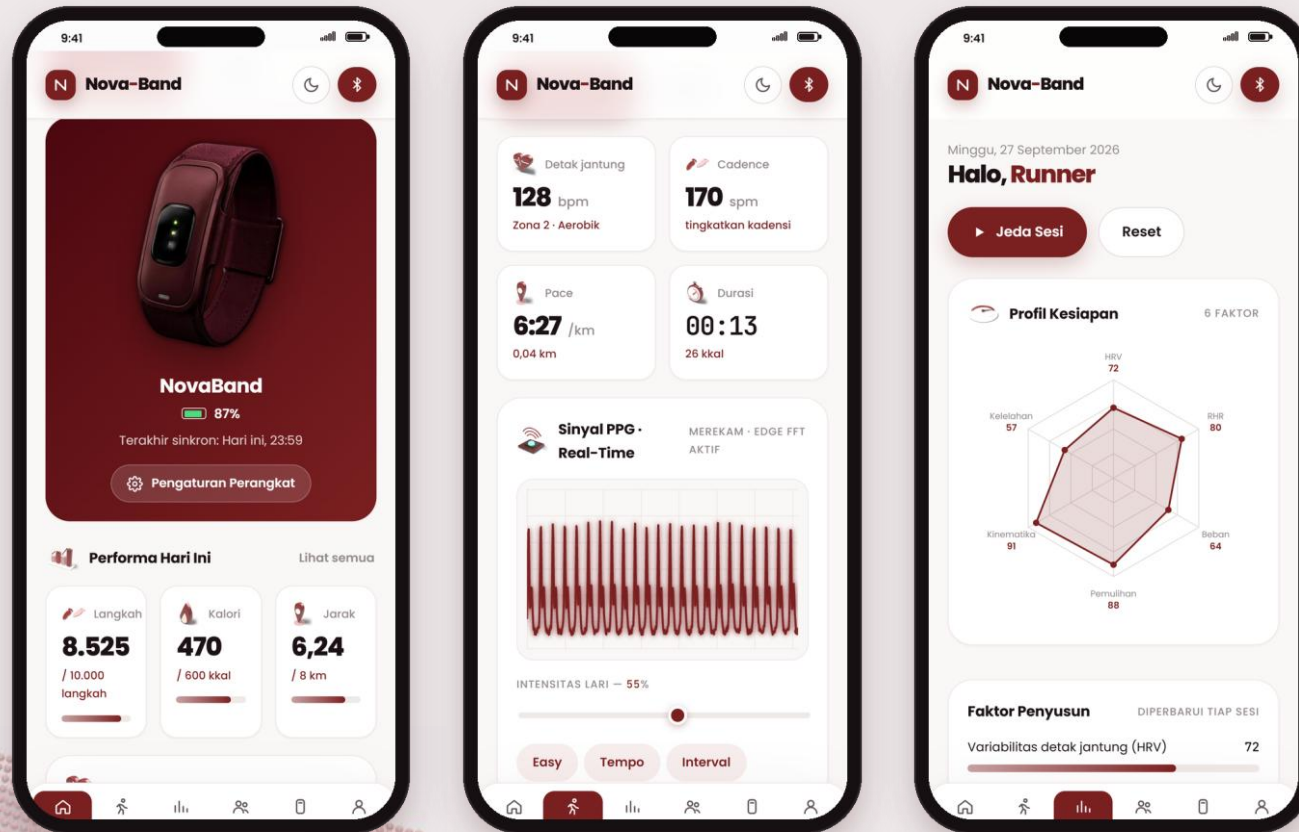
Rendered, not photographed

Blender Cycles, from the CLI. The gym, runner and props are Sketchfab models (CC-BY); the pod, its live screen and the band are ours.



Smart application

The runner app that ships with the band — real-time and explainable.



Today's performance

Steps, calories and distance against daily goals.



Live PPG signal

Filtered on the band and streamed while you run.



Readiness profile

Six factors behind one daily score.



Early warning

Alerts when a safe limit is crossed.

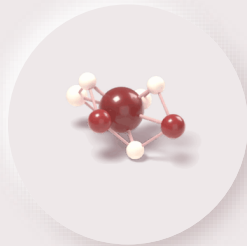
Readiness score & **early warning**

Physiology and movement, read together — so every run stays inside safe limits.



Read

Heart rate from PPG; cadence and load from the IMU.



Analyse

Six factors combined into one Readiness Score.



Warn

Alert runner and supervisor when a limit is crossed.



85

Ready to train

THE SIX FACTORS · DEMO APP

Heart-rate variability

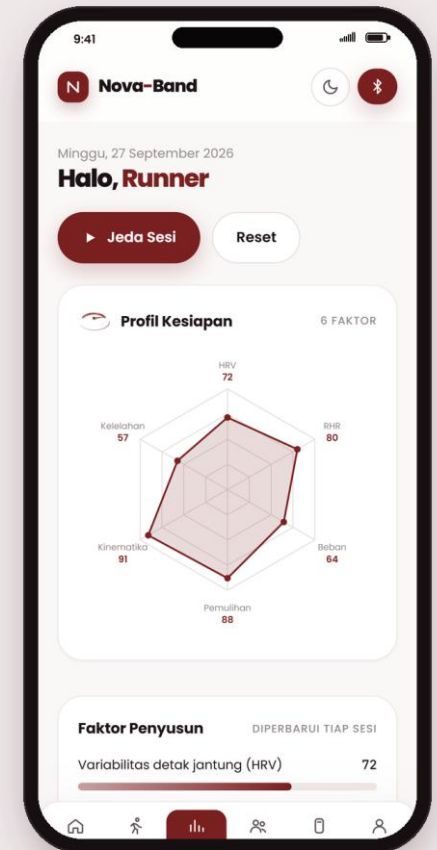
Resting heart rate

Acute : chronic load

Recovery quality

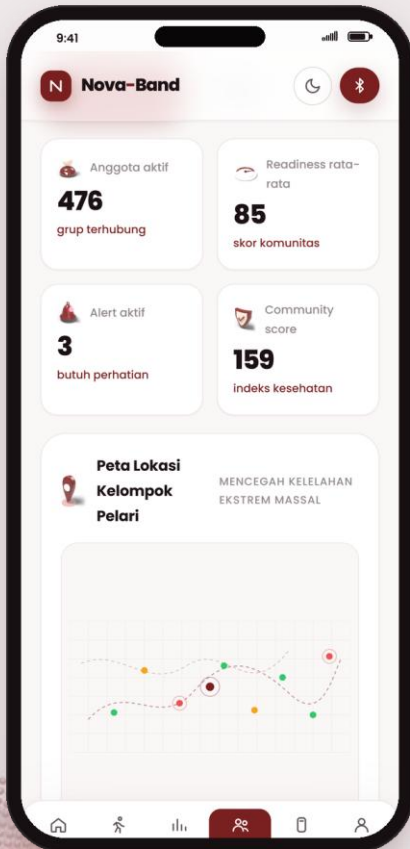
Kinematic consistency

Accumulated fatigue



Community dashboard

The Shared Safety Net for event supervisors and running groups.



Group map

Every runner's position and status on one map.



Member status

Heart rate, readiness and distance for each runner.



Simultaneous alerts

The supervisor is notified at the same moment as the runner.

476

runners connected

demo data

3

active alerts

demo data

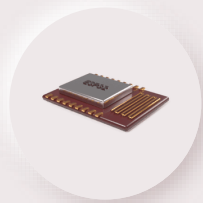
Method & timeline

Research & Development with a quantitative experimental approach.

**1**

Design Stage

- Ergonomic casing designed and 3D-printed
- Upper-arm placement to damp shocks
- PPG, IMU and ESP32 selected

**2**

Integration Stage

- Mini PCB, Li-Po and step-down assembled
- Sensor fusion + FFT programmed in C++
- Bluetooth link to app and IoT dashboard

**3**

Validation Stage

- MAPE vs high-speed motion capture
- SUS usability test, 60 respondents
- Findings feed the next design iteration

R&D · quantitative experimental

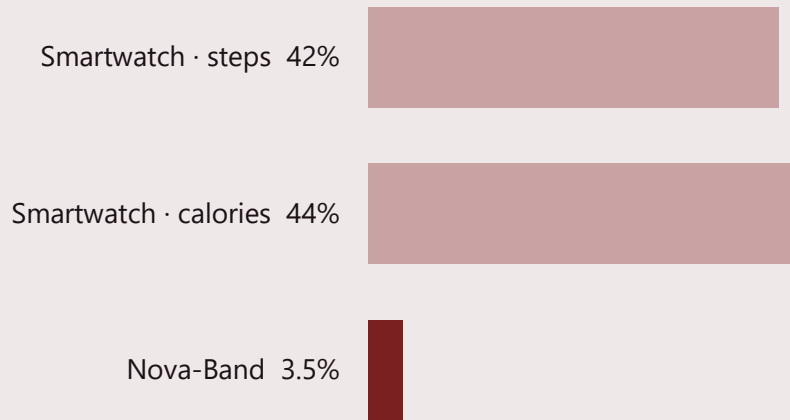
MAPE vs high-speed motion capture

SUS · 60 respondents

Result & discussion

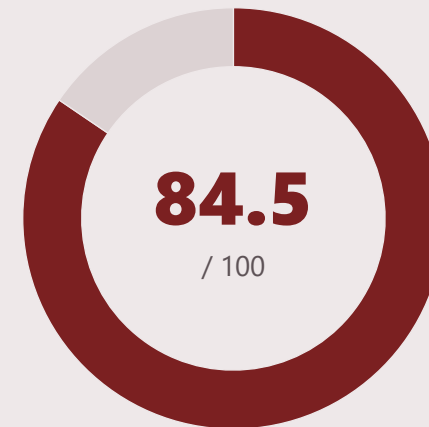
Accurate in the field, comfortable to wear.

Hardware accuracy · MAPE (%)



Nova-Band: 2.5–3.5% — below the 5% limit for professional sports instruments. Consumer values are the upper bounds of reported ranges.

Usability · SUS



Grade A · Acceptable

60 respondents · 10 SUS statements · above the 68 threshold



TESTING DOCUMENTATION

Market opportunity

Running events are where group safety matters most.

**1,818****TAM · Indonesia**

running events held nationwide in 2025

**302****SAM · Central Java**

running events in the province

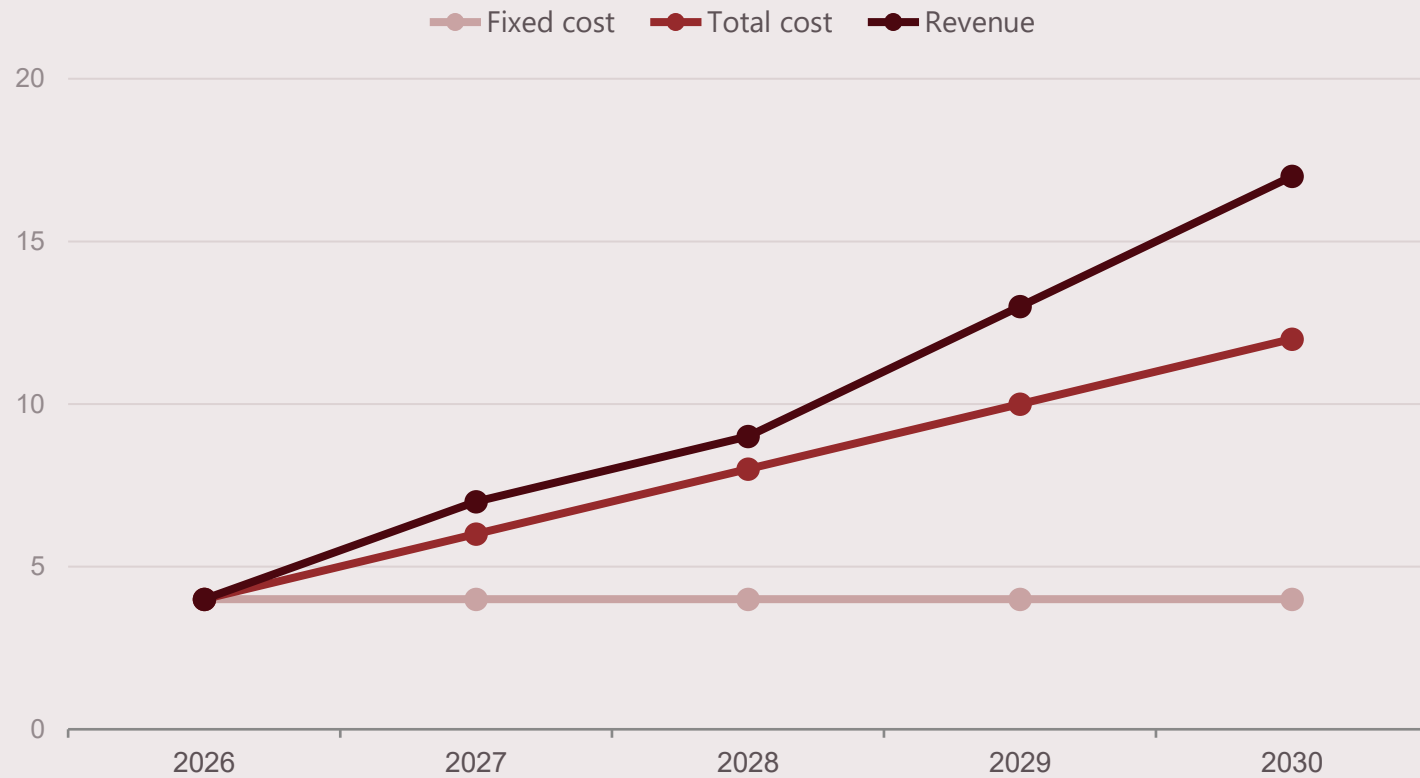
**18****SOM · Surakarta**

running events across the city

Source: 2025 running-event data, verified by the team.

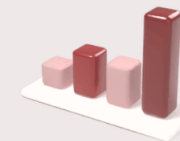
Business projection

Fixed cost, total cost and revenue, 2026–2030.



Revenue above total cost from 2027

The gap widens every year, to 17 vs 12 by 2030.



Fixed cost held flat

Values read from the chart in the ISIF deck; units as in the source.

Our partnership

Collaborators supporting research, validation and testing.



REKA Mersif

Collaborator



SMA Negeri 1 Surakarta

Research team



Universitas Sebelas Maret

Collaborator



**Mersif Academy · Enuma
Technology**

Collaborators



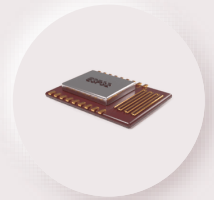
SMA Negeri 5 Yogyakarta

ISIF 2026 partner school



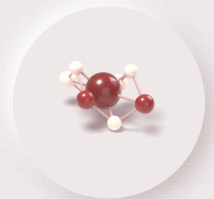
Partnership · ISIF 2026 Built by students of SMA Negeri 1 Surakarta together with university, industry and school partners.

Conclusion and **copyright**



System integration

PPG (heart rate) and IMU (body movement) on an ESP32, worn on the upper arm, collect physiological and motion data simultaneously.



AI analysis

AI processes activity patterns and physiological responses together, in real time, during training.



Impact & final result

An accurate, practical, ergonomic and data-driven wearable for real-time runner performance monitoring.

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3D asset credits

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Thank you

Monitor. Analyze. Improve.

Real-time physiological monitoring for safer running — one band, one community at a time.

TEAM NOVA-BAND

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